The field equations in a Mathematica notebook and in the ChamPBH pipeline: an audit, and a comparison of histories with common thermodynamics

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Abstract

A companion report (`.documents/peakewsurf-kicking-function`) compared the kicking function $\Sigma(T_J) = 1 - 3w(T_J)$ in the Mathematica notebook `PeakEWsurf.nb` with the one in the ChamPBH pipeline. Here we check the rest of the system. We compare the field equation term by term, together with the friction term, the law for the Jordan-frame temperature, the radiation density, the matter fraction, the potential, the parameters and the initial data. We then integrate the same histories with both codes, first giving both the pipeline's thermodynamics and then giving both the notebook's.

The two codes solve the same system. Mathematica's `Simplify` reduces the difference between each code's field equation and the equation derived from the Friedmann and Raychaudhuri equations to zero. The only structural difference is that the notebook computes ρ_r from $g_\rho(T_J)T_J^4$, while the pipeline integrates the continuity equation with the tabulated w. These differ by up to 1.6%, because the table is not exactly consistent with g_ρ and g_s, but the difference has no effect on the field. In this regime $V/\rho_r \leq 10^{-49}$ and $f_m \leq 4\times 10^{-10}$, so the equations for $y = \mathrm{d}\phi/\mathrm{d}N$ and $\ln T_J$ depend only on $\Sigma(T_J)$ and on $\gamma(T_J) = \mathrm{d}\ln g_s/3\,\mathrm{d}\ln T_J$.

With common thermodynamics the histories agree to the accuracy of the integrators. With the pipeline's Σ, γ and f_m in both codes, the notebook's `NDSolve` and the pipeline's Radau integrator agree to $|\Delta \ln T_J| \leq 7.5\times 10^{-7}$ over 39 e-folds at all 13 couplings tested. The difference is the pipeline's own truncation error; with its tolerance tightened to 10^{-10} it falls to 7×10^{-9}. Every T_J turning point and every end point agrees, on both sides of both capture thresholds. Run the other way, the pipeline's integrator reproduces the unmodified notebook to 3×10^{-5}. The two integrators place each electroweak-capture threshold within the same 3×10^{-7} in β: $\beta_c \in (3.0028522, 3.0028525]$ with the notebook's thermodynamics and $(3.0194601, 3.0194604]$ with the pipeline's. Apart from Σ, the codes differ only through the notebook's $|\cdot|$ in the temperature law. The Saikawa–Shirai fit has $\mathrm{d}g_s/\mathrm{d}T < 0$ between 0.7 and 3.5 TeV, where the $|\cdot|$ delays the history by 3.2×10^{-4} e-folds.

One result of the companion report is wrong. The notebook's $\phi_i = 100\,M_\mathrm{P}$ history is a numerical artefact. In the notebook, $\rho_r \propto (e^{\beta\phi}/M_\mathrm{P})^4$ is formed in machine arithmetic and leaves the machine range once $\beta\phi \gtrsim 220$. The true history does not depend on ϕ_i, which only sets when ϕ reaches $0.1\,M_\mathrm{P}$.

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1 The two systems, term by term

We use the notation of the companion report, with $M_{\rm P} = 1$: $y = \mathrm{d}\phi/\mathrm{d}N$, N the Einstein-frame e-fold number, $A = e^{\beta\phi}$, ρ the Einstein-frame radiation density, $f = f_m = \rho_m/\rho$, $\Sigma = 1 - 3w$ and $D = \rho(1 + f) + V$. The notebook was read from its input cells by a Mathematica 15.0.1 kernel (SHA-256 in the appendix); the pipeline from `ComputeTargets/ScalarModel.py` and `ComputeTargets/Policies/PotentialDerivativePolicy.py` at commit `5ae2b65`.

1.1 Field equation

In e-folds the Klein–Gordon equation $\ddot{\phi} + 3H\dot{\phi} + V' = -\beta\rho(\Sigma + f)$ becomes

$$y' = -\Big(3 + \frac{\dot{H}}{H^2}\Big)y - \frac{V' + \beta\rho(\Sigma + f)}{H^2}, \qquad 3H^2 = \frac{D}{1 - y^2/6}, \qquad \dot{H} = -\tfrac{1}{2}\big(\rho(1+w) + \rho f + H^2 y^2\big). \tag{1}$$

The two codes split this as follows.

Term	Notebook (cell 3)	Pipeline (`ODEPolicy`)
force and kick	`rhsPhi`: $-3(1 - \frac{y^2}{6})\dfrac{V' + \beta\rho(\Sigma + f)}{D}$	$-V'/H^2 - 3E\beta\dfrac{\Sigma + f}{1 + f}$, $E = 1 - \frac{y^2}{6} - \frac{V}{3H^2}$
friction	$-(1 - \frac{y^2}{6})\Big(1 + \dfrac{4V + (\Sigma + f)\rho}{2D}\Big)y$	$-\big(G(1 + \frac{R}{2}) + \frac{V}{6H^2}(4 - R)\big)y$, $G = 1 - \frac{y^2}{6}$, $R = \frac{\Sigma + f}{1 + f}$

`scripts/symbolic_checks.wls` gives `Simplify` the difference between Eq. (1) and each code's right-hand side. Both differences reduce to 0.

1.2 Temperature, radiation and matter

Jordan-frame temperature. Both codes use entropy conservation in the Jordan frame,

$$\frac{\mathrm{d}\ln T_J}{\mathrm{d}N} = -\frac{1 + \beta y}{1 + \gamma(T_J)}, \qquad \gamma = \frac{1}{3}\frac{\mathrm{d}\ln g_s}{\mathrm{d}\ln T_J}. \tag{2}$$

The notebook writes $\gamma = \frac{1}{3}|T_J g_s'/g_s|$. The pipeline has no $|\cdot|$. The two agree only where g_s increases with temperature (§2).

Radiation density. The notebook does not integrate ρ. It evaluates $\rho = A^4(\pi^2/30)\, g_\rho(T_J)\, T_J^4$ at each step (`RhoRadA`). The pipeline integrates $\mathrm{d}\ln\rho/\mathrm{d}N = \Sigma - 4 + \Sigma\beta y$, which is the Jordan-frame continuity equation for the tabulated w (checked symbolically, as is the equation for f). Along a history, Eq. (2) gives $\mathrm{d}\ln\tilde\rho/\mathrm{d}\ln T_J = (4-\Sigma)(1+\gamma)$ for the Jordan-frame density $\tilde\rho = A^{-4}\rho$, which depends on T_J alone. So the pipeline's $\tilde\rho$ is a function of T_J, path-independent even for histories that turn round, and differs from the notebook's by

$$\Delta(T_J) \equiv \ln\frac{\tilde\rho_{\rm pipe}}{(\pi^2/30)g_\rho T_J^4} = \int_{\ln T_J}^{\ln T_i}\Big[\Big(4+\frac{\mathrm{d}\ln g_\rho}{\mathrm{d}\ln T}\Big) - (4-\Sigma)(1+\gamma)\Big]\mathrm{d}\ln T. \tag{3}$$

Δ vanishes if w, g_ρ and g_s are mutually consistent, since then $1+w = \frac{4}{3}g_s/g_\rho$ and $\mathrm{d}p/\mathrm{d}T = s$. The pipeline's table is not exactly consistent with its g_ρ and g_s. Starting at $T_i = 2\times10^4\,\mathrm{GeV}$, we find $\Delta = +1.9\times10^{-3}$ at 100 GeV and -1.6×10^{-2} at 20 GeV. It is -1.4×10^{-2} at 1.64 GeV and -2.1×10^{-3} at 10 MeV. With the notebook's own table the values are similar, reaching -1.8×10^{-2} at 20 GeV.

Matter fraction. The notebook uses the closed form of the companion report (§3 there), with its $(1+w)$ inverted. The pipeline integrates $\mathrm{d}\ln f/\mathrm{d}N = (1-\Sigma)(1+\beta y)$. By the argument above this makes f a function of T_J alone, $f = \rho_{m0}(1+z(T_J))^3/\tilde\rho_{\rm pipe}(T_J)$.

1.3 Potential, coupling, parameters and initial data

Both codes use $A = e^{\beta\phi/M_{\rm P}}$ and $V = \Lambda^4 e^{M/\phi}$ (`ExponentialCoupling`, and `ExponentialPotential` with $n=1$). We give the pipeline the notebook's parameters: $M = 10^{-2}M_{\rm P}$, $\Lambda = 2.16\,\mathrm{meV}$ (the notebook's $v_0 = 2.16\times10^{-12}\,\mathrm{GeV}/M_{\rm P}$), $\phi_i = 10\,M_{\rm P}$ and $y_i = 0$ at $T_i = 2\times10^4\,\mathrm{GeV}$. At T_i its initial ρ is the notebook's $A^4(\pi^2/30)g_\rho T_J^4$. The two Planck masses differ, $2.43\times10^{18}\,\mathrm{GeV}$ in the notebook and $2.436\times10^{18}\,\mathrm{GeV}$ in the pipeline. Since ϕ and M are given in units of $M_{\rm P}$ and Λ and T_J in GeV, only the ratios V/ρ and f enter Eq. (1), and $M_{\rm P}$ cancels from both. The notebook stops when ϕ falls to $0.1\,M_{\rm P}$; the pipeline stops at a temperature. We therefore integrate the pipeline beyond that point and cut its history where $\phi = 0.1\,M_{\rm P}$ on its dense output.

1.4 What the dynamics depends on

Along every history integrated here, $V/\rho \le 10^{-49.7}$ and $f \le 4.3\times10^{-10}$. The companion report also found $f/\Sigma \le 3.6\times10^{-4}$ wherever Σ is appreciable. To this accuracy Eq. (1) becomes

$$y' = -\Big(1-\frac{y^2}{6}\Big)\Big[3\beta\Sigma(T_J) + \Big(1+\frac{\Sigma(T_J)}{2}\Big)y\Big], \tag{4}$$

and together with Eq. (2) the pair $(y, \ln T_J)$ evolves on its own. Neither ϕ nor the normalisation of ρ appears. Two things follow.

1. $\Delta(T_J)$, the matter fraction and the Planck mass all drop out of the dynamics here. The only inputs that can make the two codes' histories differ are $\Sigma(T_J)$ and $\gamma(T_J)$.

2. The history of (y, T_J) does not depend on ϕ_i. A larger ϕ_i only lets the same history run for longer before ϕ reaches $0.1\,M_{\rm P}$. Setting $V = 0$ in the notebook (variant `commonV0`) changes its $\beta = 3.015$ history by less than 10^{-12} in $\ln T_J$.

Conclusion 1. *The notebook and the pipeline integrate the same equations. In the regime of these runs they differ only through the functions $\Sigma(T_J)$ and $\gamma(T_J)$ they are given.*

2 The thermodynamic inputs

Both codes take g_ρ and g_s from the Saikawa–Shirai fits (arXiv:1803.01038, App. C), with identical coefficients. The notebook interpolates them linearly in T on a 10^{-4}-dex grid. The pipeline splines them at 250 points per decade. The companion report compared their Σ. Here we compare the other inputs, on a grid of 2000 points per decade from 10^{-5} to 2×10^4 GeV (`scripts/thermo_compare.py`, `data/logs/thermo_compare.txt`).

- g_ρ and g_s agree to 2.0×10^{-4} and 3.5×10^{-4}. Both maxima are at 0.12 GeV, where the fit switches between its low- and high-temperature branches, which do not quite meet.
- γ agrees to 2.4×10^{-6} between 2 and 20 GeV, and to 1.2×10^{-5} below 1 MeV. Near the branch join at 0.12 GeV the two differ by up to 0.016 in a narrow band, because the codes smooth the join differently.
- *The fitted g_s falls with temperature between 702 and* 3548 GeV, in both codes. The fit is not monotonic there; in the Standard Model g_s is. In that band $\mathrm{d}g_s/\mathrm{d}\ln T$ reaches -0.049, so the notebook's $|\cdot|$ gives γ the wrong sign there, a change of up to 3.1×10^{-4}.

At a fixed Jordan-frame e-fold number N_J ($\mathrm{d}N_J = (1+\beta y)\,\mathrm{d}N$), a change $\delta\gamma$ shifts $\ln T_J$ by $\int \delta\gamma\,\mathrm{d}\ln T/(1+\gamma)$, since $\mathrm{d}\ln T_J/\mathrm{d}N_J = -1/(1+\gamma)$. Between 1.64 GeV and T_i the notebook's γ gives a total shift of $+3.15 \times 10^{-4}$ relative to the pipeline's. The $|\cdot|$ accounts for $+3.21 \times 10^{-4}$ of this, and the difference in interpolation for -3.5×10^{-6}. The branch join at 0.12 GeV would add 1.1×10^{-4}, but no history here gets that cold. In the Einstein-frame histories the shift appears as a delay: $|\Delta \ln T_J|$ reaches 3.2×10^{-4} where T_J falls freely and is smaller while it surfs.

3 Histories with common thermodynamics

3.1 Method

Notebook. `scripts/nb_common.wls` evaluates input cells 1–3 and runs the notebook's `NDSolve` with its own options (`StiffnessSwitching`, accuracy and precision goals 10, `MaxStepSize` 10^{-3}, stop at $\phi = 0.1\,M_\mathrm{P}$). Only the thermodynamic inputs are changed:

`nb`	unchanged: the notebook's Σ, γ (with $\lvert\cdot\rvert$) and f_m
`pipeSig`	the pipeline's Σ, the notebook's γ and f_m (the companion report's variant `pipe`)
`common`	the pipeline's Σ, γ and $f_m(T_J)$

The pipeline's functions were evaluated by its own code at 2000 points per decade and passed to `NDSolve` as cubic interpolants in $\ln T_J$. The variant `nb` keeps the notebook's γ inline in the T_J equation, exactly as the notebook does.

Pipeline. `scripts/pipe_runs.py` runs the pipeline's own `ODEPolicy`, `ODERHS` and `integrate_scalar_history`, the code path used in production, with its default step control (rtol = atol = 10^{-8}). It builds the initial state as `compute_scalar_model` does. It needs neither Ray nor a datastore; this venv lacks `datastorekit`, which is replaced by a one-class stub (`scripts/dkstub.py`). The cosmology is `QCD_Cosmology` with `Planck2018`, the production configuration (variant `pipe`). In variant `nbexact` it is wrapped so that it returns the notebook's functions instead: Σ from 20 000 samples per decade of the notebook's own interpolant, and γ reproduced exactly from the nodes of the notebook's linear g_s interpolant, with its $|\cdot|$.

Couplings. Thirteen values: the companion report's nine (2.990–3.030) and a pair straddling each threshold (3.0028, 3.0029; 3.0194, 3.0195).

3.2 Results

	Notebook's thermodynamics			Pipeline's thermodynamics		
β	first min	N (nb / pipe)	N_{end}	first min	N (nb / pipe)	N_{end}
2.9900	QCD 1.6257	29.16196 / 29.16193	38.93334	QCD 1.6257	22.911205 / 22.911204	38.93343
2.9950	QCD 1.6308	32.86851 / 32.86848	38.98169	QCD 1.6307	23.808185 / 23.808184	38.97972
3.0000	none	–	36.5567	QCD 1.6357	24.972510 / 24.972510	39.02598
3.0028	none	–	35.74573	QCD 1.6385	25.802463 / 25.802462	39.05177
3.0029	EW 50.880	14.6017 / 14.6010	35.74044	QCD 1.6386	25.835263 / 25.835263	39.05268
3.0050	EW 51.315	11.30190 / 11.30190	35.67820	QCD 1.6407	26.586612 / 26.586612	39.07184
3.0100	EW 52.057	10.28088 / 10.28088	35.65433	QCD 1.6457	29.092004 / 29.092004	39.11759
3.0150	EW 52.681	9.83646 / 9.83646	35.66998	QCD 1.6508	34.083555 / 34.083554	39.15223
3.0194	EW 53.179	9.57929 / 9.57929	35.69265	none	–	35.94865
3.0195	EW 53.190	9.57429 / 9.57429	35.69322	EW 49.231	14.51178 / 14.51177	35.93766
3.0200	EW 53.244	9.54975 / 9.54975	35.69608	EW 49.357	12.368522 / 12.368522	35.89539
3.0250	EW 53.768	9.33802 / 9.33802	35.72657	EW 50.153	10.478769 / 10.478769	35.82252
3.0300	EW 54.264	9.17021 / 9.17021	35.75948	EW 50.768	9.966599 / 9.966599	35.84837

Table 1: The first T_J minimum (in GeV) and the e-fold at which each code finds it, from the notebook (`nb` and `common`) and the pipeline (`nbexact` and `pipe`). N_{end}, where $\phi = 0.1\,M_{\text{P}}$, agrees in every row to the digits shown; the largest difference is 4.4×10^{-5}, at $\beta = 3.000$ with the notebook's thermodynamics. "none" means no minimum before N_{end}. The minimum temperatures agree to 6×10^{-7} (left) and 6×10^{-9} (right) in $\ln T_J$.

Table 1 and Fig. 1 give the results.

- **Pipeline's thermodynamics in both.** The notebook (`common`) and the pipeline (`pipe`) agree to $|\Delta \ln T_J| \leq 7.5 \times 10^{-7}$, $|\Delta\phi| \leq 2.5 \times 10^{-7}$ and $|\Delta y| \leq 1.3 \times 10^{-7}$ at every coupling. Every turning point appears in both, in the same order, at the same N to 10^{-6} (9×10^{-6} at $\beta = 3.0195$, the coupling closest to its threshold). This difference is exactly the pipeline's own truncation error: rerunning the pipeline at rtol $= 10^{-10}$ changes it by the same 7.6×10^{-7}. At that tolerance it lies within 7×10^{-9} of the notebook in $\ln T_J$ and 2×10^{-9} in ϕ, at every coupling (Fig. 1b, c).

- **Notebook's thermodynamics in both.** The pipeline (`nbexact`) reproduces the unmodified notebook to $|\Delta \ln T_J| \leq 3.0 \times 10^{-6}$ for the histories captured at the electroweak scale. For those that reach the QCD transition ($\beta \leq 3.000$) the bound is 3.0×10^{-5}. In this pairing the notebook's γ is piecewise constant on 10^{-4}-dex cells, so both integrators step across roughly 10^4 discontinuities per decade of temperature. That is the most likely source of the larger residual; we did not separate it further. Resampling the notebook's functions at 2000 points per decade instead (variant `nbthermo`) gives 6×10^{-4}, so the sampling must be fine.

- **Common Σ only.** The companion report's runs (`pipeSig`) differ from the pipeline by up to 3.2×10^{-4} in $\ln T_J$ at every coupling. This is a delay of 3.2×10^{-4} e-folds, seen in full where T_J falls freely and less while it surfs (Fig. 1b); the end points move by the same 3.2×10^{-4}. It comes from the notebook's $|\cdot|$ (§2), and it is dynamically negligible.

The thresholds. We bisected with the pipeline's integrator to a tolerance of 10^{-6} in β (`data/logs/pipeline_bisections.txt`). We then ran the notebook at five couplings across each pipeline bracket (`data/logs/notebook_thrnb_log.txt`, `notebook_thrcommon_log.txt`). Each row uses one set of thermodynamics in both codes.

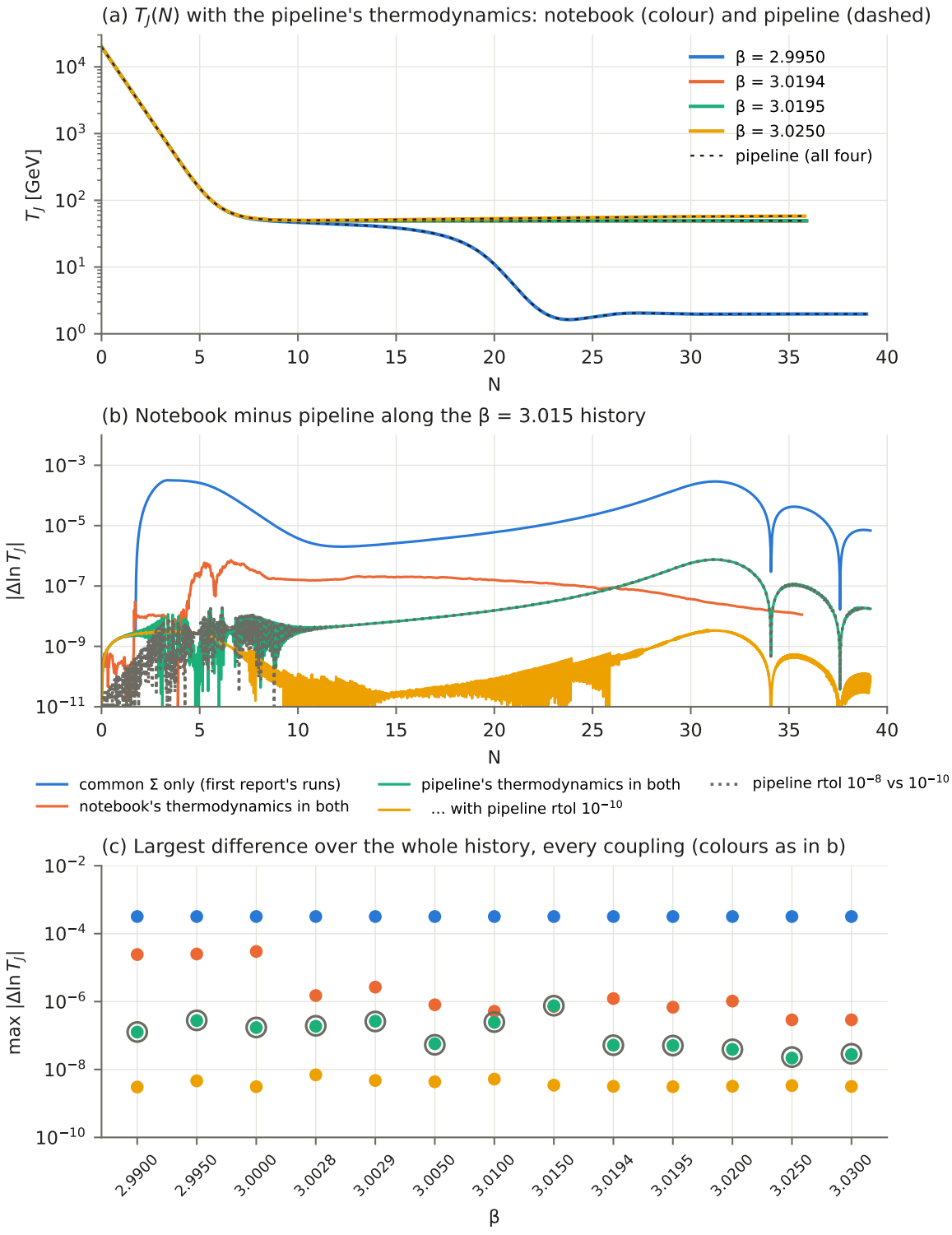


Figure 1: (a) $T_J(N)$ with the pipeline's thermodynamics, from the notebook (colour) and the pipeline (dashed, drawn over each). $\beta = 3.0194$ and 3.0195 lie either side of the electroweak-capture threshold. (b) The difference in $\ln T_J$ along the $\beta = 3.015$ history for each pairing. The dotted curve is the pipeline against itself at tolerances 10^{-8} and 10^{-10}. (c) The largest difference over each whole history. Open markers are the dotted pairing of panel (b); they ring the green markers, which they coincide with.

Thermodynamics	Pipeline integrator	Notebook integrator	Notebook (companion report)
notebook's	(3.0028522, 3.0028528]	(3.0028522, 3.0028525]	(3.002813, 3.002891)
pipeline's	(3.0194598, 3.0194604]	(3.0194601, 3.0194604]	(3.019453, 3.019531)

The two integrators place each threshold within the same 3×10^{-7} in β. The pipeline brackets are unchanged at rtol $= 10^{-10}$, and unchanged when the notebook's Σ is used with either code's γ (variants `nbsig`, `nbthermo` and `nbexact`). The companion report's column used the notebook's γ and f_m throughout. The pipeline's own fine scan near $\beta = 3$ places its edge in $(3.019, 3.020]$. That scan used another potential and ϕ_i, and its own classification of capture; neither the potential nor ϕ_i enters the (y, T_J) equations here (§1), so the agreement is expected.

Conclusion 2. *Given the same Σ and γ, the notebook and the pipeline yield the same field histories to the accuracy of their integrators, and the same capture thresholds. Every difference between them traced so far comes from Σ (companion report) or, at the level of 3×10^{-4} in $\ln T_J$, from the notebook's $|\cdot|$.*

4 Large initial fields: machine overflow in the notebook

By §1 the (y, T_J) history does not depend on ϕ_i. The pipeline confirms this: at $\beta = 3.005$ its $\phi_i = 10$, 50, 58, 60 and $100\,M_\mathrm{P}$ histories have the same turning points to all digits printed. The notebook does too, up to $\phi_i = 72\,M_\mathrm{P}$: its $\phi_i = 50$, 58, 60, 66, 68, 70 and $72\,M_\mathrm{P}$ histories reproduce its $\phi_i = 10\,M_\mathrm{P}$ one to $|\Delta \ln T_J| \leq 2.5 \times 10^{-6}$, most of them to 10^{-9}. At $\phi_i = 75$ and $100\,M_\mathrm{P}$ they do not (Fig. 2).

Cause. The notebook evaluates $\rho = (A/M_\mathrm{P})^4(\pi^2/30)g_\rho T_J^4$ with $A = e^{\beta\phi}$ (`RhoRadA`), and $(A/M_\mathrm{P})^4$ exceeds the largest machine number, 1.8×10^{308}, once $\beta\phi > \ln(1.16 \times 10^{77} \times 2.43 \times 10^{18}) = 219.8$: at $\phi = 73.1\,M_\mathrm{P}$ for $\beta = 3.005$. The sweep brackets the onset between $\beta\phi_i = 216.4$ (correct) and 225.4 (wrong), consistent with this. Mathematica then switches silently to arbitrary-precision numbers (the integrations run under `Quiet`), and some combinations underflow: in machine arithmetic $0.03 \times (4.0 \times 10^{457})^{-1}$ returns 0 with a `General::munfl` message. Evaluated one point at a time, the notebook's right-hand side is still exactly the same at $\phi = 10$ and $100\,M_\mathrm{P}$. Inside `NDSolve`, however, one e-fold from $y = -0.05$, $T_J = 200\,\mathrm{GeV}$ gives $y = -0.148691$ from $\phi = 10$ or $50\,M_\mathrm{P}$ but -0.149436 from $\phi = 100\,M_\mathrm{P}$. We did not identify which internal evaluation goes wrong. The error persists for as long as $\phi \gtrsim 73\,M_\mathrm{P}$, that is until $N \approx 87$ for $\phi_i = 100$. The spurious T_J maximum the companion report found at $N = 82.95$ (where $\phi = 74.3\,M_\mathrm{P}$) belongs to the end of this stretch.

Check. `nb_common.wls` has log-space variants (suffix `Log`) with the numerator and denominator of the right-hand side divided by ρ, and V/ρ formed from logarithms. The pipeline computes $V/3H^2$ the same way. The change is algebraically nothing, and at $\phi_i = 10$ it reproduces the notebook to ten digits. At $\phi_i = 100\,M_\mathrm{P}$ the log-space notebook agrees with the pipeline over all 150 e-folds, to $|\Delta \ln T_J| \leq 8 \times 10^{-7}$ with the notebook's thermodynamics and 3×10^{-8} with the pipeline's (Fig. 2). With the notebook's thermodynamics, T_J has a single minimum at $N = 11.30$ (51.31 GeV), the same as for $\phi_i = 10$, then rises slowly to 58.36 GeV at $N = 150$. With the pipeline's, the history is captured at the QCD transition at $N = 26.59$ (1.641 GeV), as for $\phi_i = 10$, and oscillates about 1.992 GeV.

Consequences.

1. *The companion report's "longer history" (its §4.2 and Fig. 5) is an artefact.* The true $\phi_i = 100\,M_\mathrm{P}$ histories are those just described, not an electroweak minimum at $N \approx 9$–10

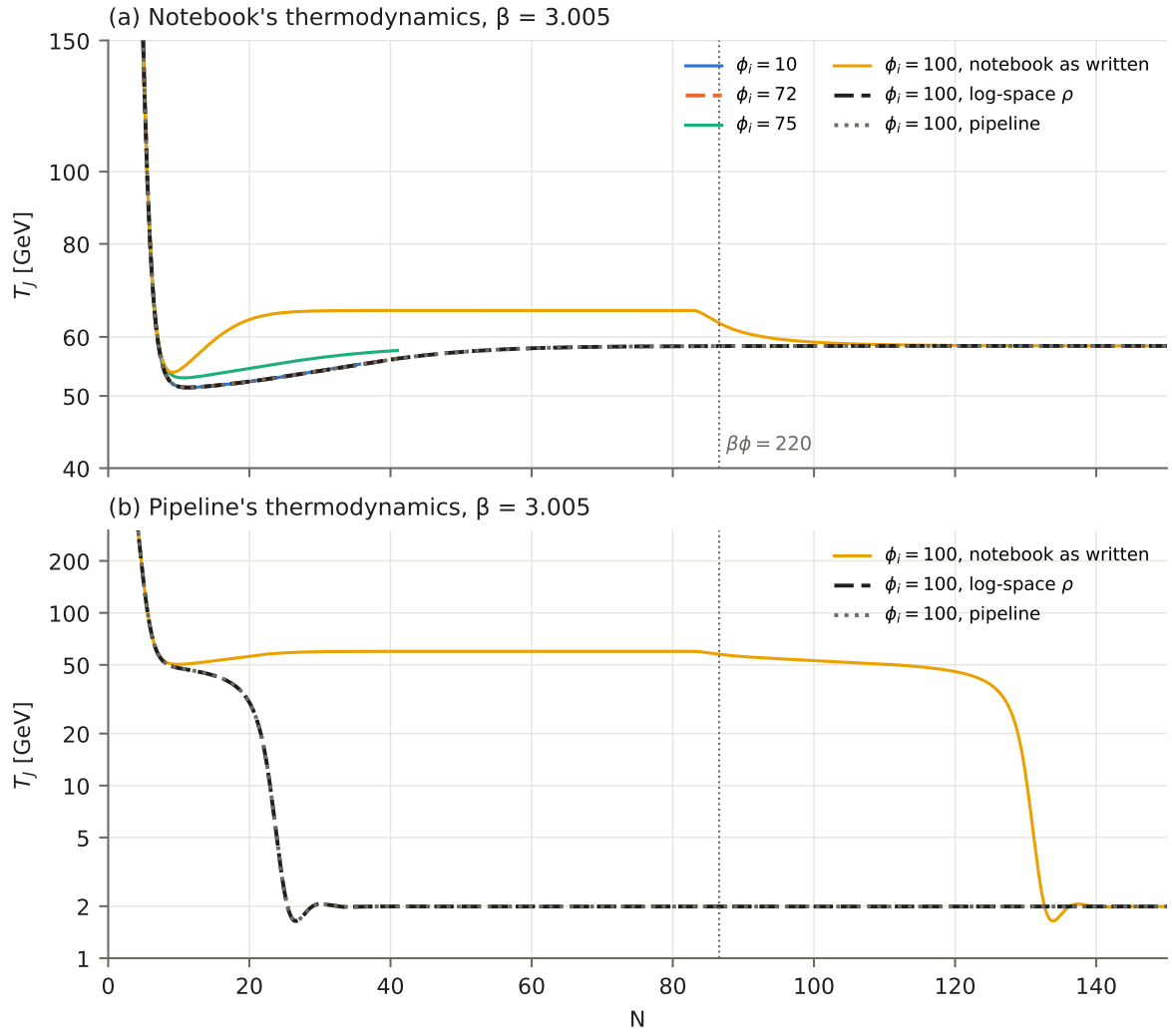


Figure 2: $T_J(N)$ at $\beta = 3.005$. (a) The notebook's thermodynamics: the notebook at $\phi_i = 10$, 72, 75 and $100\,M_{\mathrm{P}}$, the log-space notebook at $100\,M_{\mathrm{P}}$, and the pipeline at $100\,M_{\mathrm{P}}$. The $\phi_i \le 72$ curves, the log-space curve and the pipeline's lie on top of one another. (b) The pipeline's thermodynamics at $\phi_i = 100\,M_{\mathrm{P}}$. The dotted line marks where the $\phi_i = 100\,M_{\mathrm{P}}$ history falls back through $\beta\phi = 220$.

for both Σ followed by a maximum at $N \approx 83$. Its other results use $\phi_i = 10\,M_{\mathrm{P}}$ ($\beta\phi \le 30.3$) and stand.

2. The notebook's own $\phi_i = 100\,M_{\mathrm{P}}$ cells are affected in the same way. These are cells 10–11 ($\beta = 3.005$) and cell 16 ($\beta = 2.35$, $\beta\phi_i = 235$), and with them the plots drawn from those solutions (cells 12–15 and 17–22).

5 Observations not acted on

- **The pipeline's ρ_r is not $(\pi^2/30)g_\rho T_J^4$.** Because the w table is not exactly consistent with g_ρ and g_s, the integrated $\tilde{\rho}$ differs from it by $\Delta(T_J)$ (Eq. 3), up to 1.6% near 20 GeV for $T_i = 2 \times 10^4$ GeV. The field histories here do not depend on it. The pipeline's stored ρ_r and H do, and so would any regime in which V/ρ_r is not negligible. Because Δ is a function of T_J and T_i alone, it is easy to tabulate if needed.

- **The Saikawa–Shirai g_s fit is not monotonic** between 0.7 and 3.5 TeV. The effect on

these histories is 3×10^{-4} in $\ln T_J$.

- **The venv has no `datastorekit`.** `requirements.txt` pins `v0.4.0`, but `import datastorekit` fails in `./venv`, so as it stands the test suites cannot import `ComputeTargets.ScalarModel`. These runs used a stub.

6 Summary and recommendations

1. The notebook and the pipeline integrate the same field equation, friction term and temperature law (§1). Their only structural difference, ρ_r algebraic against integrated, has no effect on the field in this regime.
2. Given the same Σ and γ their histories agree to 7×10^{-9} in $\ln T_J$, and their capture thresholds agree to 3×10^{-7} in β (§3). The difference between the codes' published capture edges is therefore entirely the difference in Σ that the companion report found.
3. Apart from Σ, the notebook differs only through its $|\cdot|$ in γ, which delays the histories by 3.2×10^{-4} e-folds.
4. The notebook's $\phi_i = 100\, M_{\mathrm{P}}$ results are corrupted by machine overflow (§4).

We recommend:

- In the notebook, form the right-hand side in terms of V/ρ and V'/ρ evaluated through logarithms, as the log-space variant does, or keep $\beta\phi < 220$. Remove the Quiet around `NDSolve`, or at least catch `General::ovfl` and `General::munfl`.
- In the notebook, drop the $|\cdot|$ in γ, so that it matches the pipeline.
- Correct the companion report's §4.2 and Fig. 5 with an additive note pointing here.

Not covered. Regimes in which V/ρ_r or f_m is not negligible (late times, smaller ϕ near M, or the pipeline's reflections at the wall), where the normalisation $\Delta(T_J)$ and the matter fraction would start to matter. Temperatures below 0.12 GeV, where the two codes' γ differ near the branch join. Couplings far from $\beta = 3$.

A Reproducing these results

Versions.

- Notebook `PeakEWsurf.nb`, not under version control, SHA-256 `5bd78e6f43b6b7fc4a2eaff746263b63a66c27786eef76cea90edbdd070e92e4` (the same file as the companion report's).
- ChamPBH at commit `5ae2b65`; Mathematica 15.0.1 (macOS, ARM); Python 3.13 with NumPy 2.3.5 and SciPy 1.17.0.

Files. Everything is in `.documents/peakewsurf-field-equations/`.

- `scripts/symbolic_checks.wls`: the symbolic checks of §1.
- `scripts/nb_thermo.wls`, `nb_export_exact.wls`: export the notebook's thermodynamic functions; `thermo_compare.py`: compares them with the pipeline's, computes $\Delta(T_J)$ and writes the pipeline's functions for `nb_common.wls`.

- `scripts/nb_common.wls`: the notebook's integration with swapped inputs; `rhs_phi_dep.wls`: the overflow demonstration.
- `scripts/pipe_runs.py`, `bisect_pipe.py`, `dkstub.py`: the pipeline's integrations and bisections.
- `scripts/compare_histories.py`, `build_data.py`, `make_figures.py`: the comparison, the data products and the figures.
- `data/`: `compare.json` (every pairing, every coupling), `max_residuals.csv`, `pipeline_runs.json` (every pipeline run's turning points), downsampled histories and residuals, and the logs of every notebook run, bisection and comparison.

Commands. Run from the repository root, with `WORK` a scratch directory and `WolframKernel` set as in the companion report.

```
D=.documents/peakewsurf-field-equations/scripts
wolframscript -file $D/nb_thermo.wls $PWD/PeakEWsurf.nb $WORK
wolframscript -file $D/nb_export_exact.wls $PWD/PeakEWsurf.nb $WORK
./venv/bin/python $D/thermo_compare.py $WORK
export NB_THERMO_GRID=$WORK/nb_thermo_grid.csv NB_EXACT_DIR=$WORK
B=2.99,2.995,3.0,3.0028,3.0029,3.005,3.01,3.015
B=$B,3.0194,3.0195,3.02,3.025,3.03
for v in pipe nbexact nbthermo nbsig; do
  ./venv/bin/python $D/pipe_runs.py $WORK/pipe $B $v 10 41; done
./venv/bin/python $D/pipe_runs.py $WORK/pipe_r10 $B pipe 10 41 1e-10
./venv/bin/python $D/pipe_runs.py $WORK/pipe100 3.005 pipe 100 150   # and nbexact
wolframscript -file $D/nb_common.wls $PWD/PeakEWsurf.nb $WORK/nbrun \
    $WORK/pipe_thermo_grid.csv "{${B}}" '{"nb","pipeSig","common"}' \
    "10^-3" 10 150 scan True \
    > $WORK/nbrun/scan_log.txt
./venv/bin/python $D/bisect_pipe.py 2.995 3.005 nbexact 1e-6
./venv/bin/python $D/bisect_pipe.py 3.015 3.02 pipe 1e-6
```

The ϕ_i sweep runs `nb_common.wls` with $\phi_i \in \{50, 58, 60, 66, 68, 70, 72, 75\}$ (tags `phi`ϕ_i and `onset`ϕ_i, $N_{\max} = 41$), and with $\phi_i = 100$, $N_{\max} = 150$ for the variants `nb`, `common`, `nbLog` and `commonLog` (tag `p100long`). The pipeline's $\phi_i = 50$, 58 and 60 runs go to `WORK/pipephi`ϕ_i. The notebook threshold runs pass two more arguments, the decimals in the file names and a time limit per run: `... "10^-3" 10 150 thrnb True 7 240` with the variant `"nb"`, and the same with `"common"` and tag `thrcommon`. Then `compare_histories.py WORK`, `build_data.py WORK` and `make_figures.py`. A notebook integration takes 20–30 s for `nb`, and 1–3 min for the swapped variants, whose functions are evaluated through `?NumericQ` wrappers. The log-space variants take longer still. A pipeline integration takes under a second.